

EC466: Econometrics

Problem Set # 8: Callaway-Sant'Anna Estimator and Modern DiD Methods

Background: Consider a training program that was implemented across different cities at different times. You have panel data from five cities (E, F, G, H, I) over seven time periods (2017-2023), with the following average employment rates (in percentage points):

City	2017	2018	2019	2020	2021	2022	2023	Treatment Start
E	62.5	63.2	70.8	74.1	77.5	80.2	82.8	2019
F	61.8	62.4	63.1	71.2	74.9	78.3	81.5	2020
G	62.1	62.9	63.6	64.3	72.8	76.5	79.9	2021
H	63.3	64.0	64.7	65.4	66.1	74.2	77.8	2022
I	61.2	61.8	62.4	63.0	63.6	64.2	64.8	Never

Each city has 150 observations per period, and treatment is irreversible once adopted.

1. Group-Time Average Treatment Effects

- Define $ATT(g, t)$ precisely in the context of this staggered adoption design. What does g represent? What does t represent? What is the interpretation of $ATT(2020, 2022)$ in this application?
- Using City I (never treated) as the comparison group, calculate $ATT(2019, 2021)$ for City E "by hand" using the unconditional estimand:

$$ATT_{unc}^{nev}(g, t) = E[Y_t - Y_{g-1} | G_g = 1] - E[Y_t - Y_{g-1} | G_g = \infty]$$

Show all your work and interpret the result.

- Now calculate $ATT(2020, 2022)$ for City F using the same approach. How does this estimate compare to the one you calculated in part (b)? What might explain the difference?

2. Alternative Comparison Groups

- (a) Explain the difference between using “never-treated” units versus “not-yet-treated” units as a comparison group. Write down the parallel trends assumption for each case.
- (b) Using City H as a “not-yet-treated” comparison group, calculate $ATT(2020, 2021)$ for City F. How does this estimate differ from using City I (never-treated) as the comparison? Which comparison do you find more credible and why?
- (c) Suppose you wanted to estimate $ATT(2019, 2023)$ for City E. Which cities could serve as valid “not-yet-treated” comparison groups in 2023? Explain your reasoning. What does this imply about the availability of comparison groups over time?

3. Aggregation and Summary Measures

- (a) Consider the simple average aggregation:

$$\theta_M := \frac{2}{T(T-1)} \sum_{g=2}^T \sum_{t=2}^T 1\{g \leq t\} ATT(g, t)$$

Calculate the number of group-time ATTs that would be averaged in θ_M for the data above (i.e., how many (g, t) pairs satisfy $g \leq t$). Explain why this aggregation might “overweight” earlier-treated cities.

- (b) Write down the formula for the event-study aggregation $\theta_D(e)$, where e represents periods of exposure to treatment. What is the interpretation of $\theta_D(2)$ in this application?
- (c) Compare $\theta_D(2) - \theta_D(1)$ to the difference in individual group effects $ATT(g, g+2) - ATT(g, g+1)$ for a specific group g . Under what conditions would these two measures of dynamic treatment effects be identical? When might they differ?